

FairLens AI Bias Audit Report

Audit ID: FL-2026-655881

Date: May 03, 2026

Submitted by: Manual Input User

Executive Summary

This audit assessed an AI-powered loan approval system for bias, discrimination, and regulatory compliance. The system received an overall risk rating of **HIGH RISK** ■ with a composite fairness score of **73.863636363636/100**.

The model demonstrates acceptable fairness characteristics and is suitable for deployment with ongoing monitoring.

Regulatory Context

This audit evaluates compliance with:

- **Equal Credit Opportunity Act (ECOA):** Prohibits discrimination in credit decisions
- **EU AI Act Article 9:** Classifies credit scoring as high-risk AI requiring strict oversight
- **Fair Credit Reporting Act (FCRA):** Regulates use of consumer credit information
- **GDPR Article 5:** Requires data minimization and privacy protection
- **Federal Reserve SR 11-7:** Model risk management guidance

1. Data Bias Risk Assessment

COMPLIANT ■

No direct use of protected attributes detected in model features.

■ *No Proxy Variables Detected*

No obvious proxy variables found in the feature set.

2. Discrimination Detection Results

■ *Disparate Impact Analysis*

Disparate Impact Ratio: 0.739

Statistical Parity Difference: -0.080

Equal Opportunity Difference: -0.080

Interpretation:

DISCRIMINATION DETECTED: Disparate Impact ratio of 0.739 is below the 0.8 threshold (4/5 rule).
Female approval rate: 22.7%, Male approval rate: 30.8%

Detailed Metrics:

- Privileged group selection rate: 30.8%
- Unprivileged group selection rate: 22.7%
- Total samples analyzed: 35
- Privileged samples: 13
- Unprivileged samples: 22

3. Accountability & Governance

Governance Maturity Level: Defined (Level 2)

Maturity Score: 60/100

Risk Level: ■ MEDIUM

RACI Matrix

4. Data Privacy Compliance

5. Remediation Recommendations

Audit Trail

Audit ID: FL-2026-655881

Audit Date: 2026-05-03 10:53:51 UTC

Submitted By: Manual Input User

Model Version: 1.0.0

Next Review Date: 2026-08-03

Composite Risk Score: 73.86363636363636/100

Overall Risk Level: HIGH RISK ■

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